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ACME PETROCHEMICAL DEMO COMPLEX - CORPORATE PROCESS SAFETY STANDARD

DOCUMENT CODE: STD-PHA-001 | REVISION: R1 | MANDATORY GOVERNANCE PROCEDURE

TITLE: PROCESS HAZARD ANALYSIS (PHA), 5x5 RISK RANKING MATRIX & HAZOP STUDY PROCEDURE

1. PURPOSE & SCOPE

This standard defines the mandatory corporate methodology for conducting Hazard and Operability (HAZOP) studies,

2. CORPORATE 5x5 RISK RANKING MATRIX

Severity Categories (Consequence to Personnel, Environment, and Plant Assets):

- Severity 5 (Catastrophic): Uncontained runaway reaction, vessel rupture, or toxic/flammable release with potential for major loss of life or major environmental damage.
- Severity 4 (Major): Major loss of primary containment (LOPC), permanent disability, or \$5M - \$50M plant damage.
- Severity 3 (Serious): Lost-time injury, localized fire/spill, or PSV lift to closed relief system (\$500k - \$5M).
- Severity 2 (Minor): Recordable first-aid event or minor equipment outage (\$50k - \$500k).
- Severity 1 (Negligible): Process upset within design envelope (< \$50k).

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Likelihood Categories (Initiating Event Frequency):

- Likelihood E (Frequent): > 1 event per year ($> 10^{-1}$ /yr)
- Likelihood D (Probable): 10^{-2} to 10^{-1} events/yr
- Likelihood C (Occasional): 10^{-3} to 10^{-2} events/yr
- Likelihood B (Remote): 10^{-4} to 10^{-3} events/yr
- Likelihood A (Improbable): $< 10^{-4}$ events/yr

Risk Tiers & Mandatory Action Thresholds:

- TIER 1 - EXTREME / UNACCEPTABLE RISK (Red: 5C, 5D, 5E, 4D, 4E): Immediate engineering redesign and at least one
- TIER 2 - HIGH / ALARP RISK (Orange: 5B, 4C, 3D, 3E): Requires verified SIL-1/SIL-2 SIS interlock and mechanical
- TIER 3 - MEDIUM RISK (Yellow: 5A, 4B, 3C, 2D): Manage with independent BPCS alarms and standard operating procedure
- TIER 4 - LOW / ACCEPTABLE RISK (Green: 4A, 3A, 3B, 2A-2C, 1A-1E): Acceptable with routine maintenance.

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Standard HAZOP Guideword Matrix:

- HIGH TEMPERATURE (More Temperature): Exothermic runaway, loss of cooling medium (E-2307 TV-1302 failure), or ex
- HIGH PRESSURE (More Pressure): Blocked vapor/liquid outlet (LV-1301 closed), loss of vacuum (X-2301 trip), or t
- LOW FLOW / NO FLOW: Recirculation pump P-2304A/B trip or plugged exchanger E-2307 tubes causing localized hot s
- HIGH CONCENTRATION (More Composition): Excess CHP accumulation > 2.5 wt% in D-2304 prior to acid initiation.

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Node 2300-01: Preflash Column V-2301 & Vacuum Package X-2301

- Deviation: High Temperature / Loss of Vacuum (> 105 degC in V-2301 sump containing 82 wt% CHP)
- Unmitigated Consequence: Thermal auto-decomposition of 82 wt% CHP (SADT 75 degC), rapid pressure rise (Severity 4)
- Independent Protection Layers (IPLs):
 1. Interlock I-2301 (SIL-2): TAAH-0402 >= 105 degC or PAHH-0401 >= 75 mmHgA trips reboiler steam XV-0403 and isolates V-2301
 2. Mechanical Relief: PSV-2301 (Set 3.5 kg/cm2g, Orifice 3K4) relieves to Knock-Out Drum D-2320.

Node 2300-02: Decomposer Reactor D-2304 & Circulation Cooler E-2307

- Deviation: High Temperature (> 115 degC) / Loss of E-2307 Cooling or Excess Acid Catalyst
- Unmitigated Consequence: Exponential exothermic CHP cleavage runaway (-1580 kJ/kg), two-phase overpressure exceeding design pressure
- Independent Protection Layers (IPLs):
 1. BPCS Temperature Control Loop TIC-1302 modulating E-2307 cooling water valve TV-1302 (Fails Open FO) + High Pressure Alarm
 2. SIS Interlock I-2304 (SIL-2, PFDavg = 0.004): 2oo3 Voting TXSHH-1301 >= 115.0 degC or PAHH-1301 >= 5.5 kg/cm2g
 3. Mechanical Overpressure Protection: Dual Balanced-Bellows Relief Valves PSV-2304A/B (Set 11.0 kg/cm2g, Orifice 3K4)

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- Mandatory PSI Verification Rule: Prior to closing any Tier 1 or Tier 2 HAZOP node, the facilitator must verify
- Open Action Item HAZOP-ACT-2304-01: Resolve design pressure discrepancy on D-2304 between DS-D2304 (11.0 kg/cm²